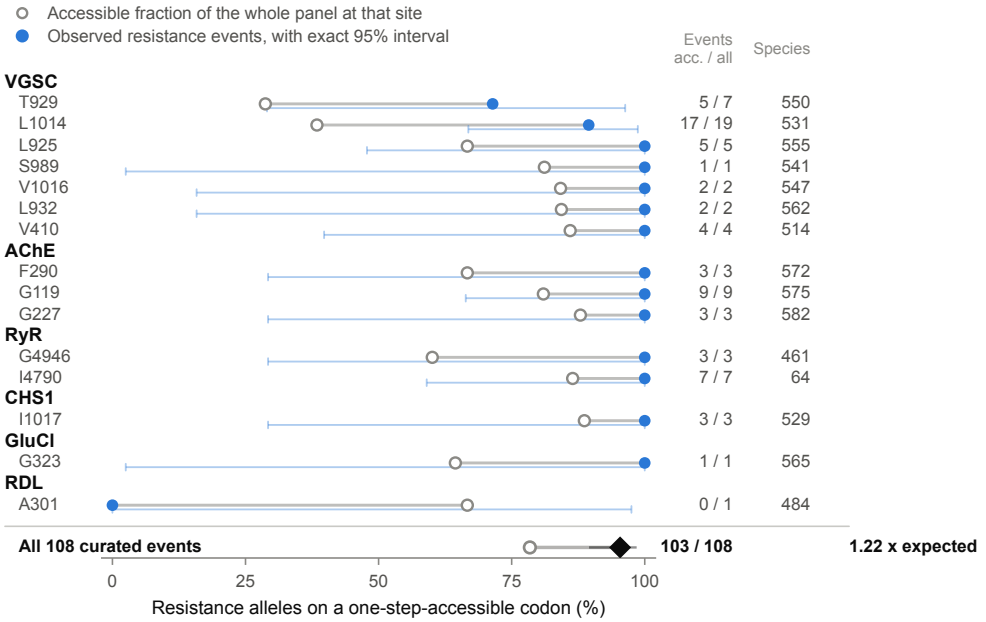
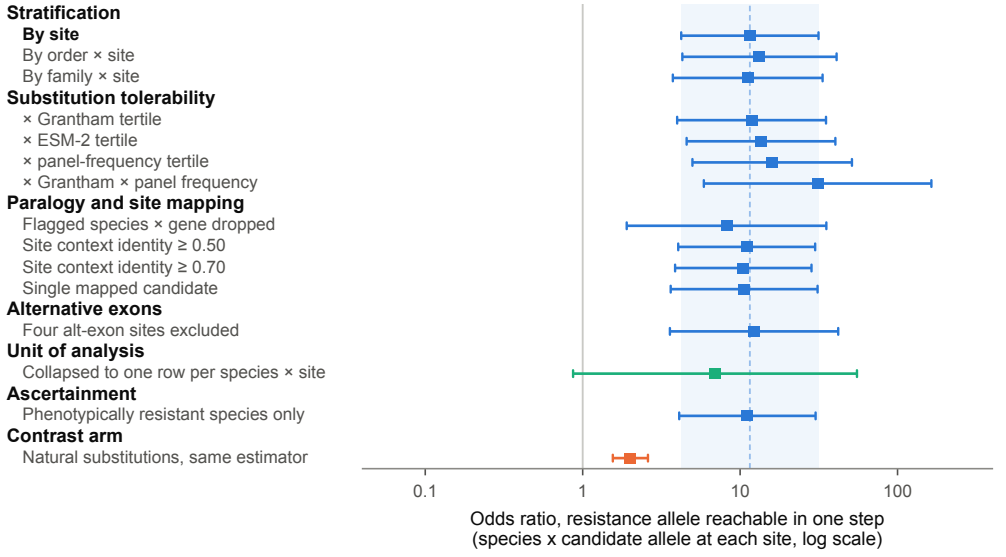
a


Accessible fraction of the whole panel at that site
Observed resistance events, with exact 95% interval
Events acc. / all
Species
VGSC
T929 5 / 7 550
L1014 17 / 19 531
L925 5 / 5 555
S989 1 / 1 541
V1016 2 / 2 547
L932 2 / 2 562
V410 4 / 4 514
AChE
F290 3 / 3 572
G119 9 / 9 575
G227 3 / 3 582
RyR
G4946 3 / 3 461
I4790 7 / 7 64
CHS1
I1017 3 / 3 529
GluCl
G323 1 / 1 565
RDL
A301 0 / 1 484
All 108 curated events
103 / 108
1.22 x expected
0
25
50
75
100
Resistance alleles on a one-step-accessible codon (%)

b


Stratification
By site
By order × site
By family × site
Substitution tolerability
× Grantham tertile
× ESM-2 tertile
× panel-frequency tertile
× Grantham × panel frequency
Paralogy and site mapping
Flagged species × gene dropped
Site context identity ≥ 0.50
Site context identity ≥ 0.70
Single mapped candidate
Alternative exons
Four alt-exon sites excluded
Unit of analysis
Collapsed to one row per species × site
Ascertainment
Phenotypically resistant species only
Contrast arm
Natural substitutions, same estimator
0.1
1
10
100
Odds ratio, resistance allele reachable in one step
(species x candidate allele at each site, log scale)